

What each microbiome assay can measure

Direct measurement does not imply freedom from bias or confounding

	Community structure	Species / strain	Gene potential	Gene activity	Metabolites	Absolute load
16S amplicon	Direct	Limited	Predicted	No	No	No
Shotgun metagenome	Direct	Strong	Direct	No	No	Relative
Metatranscriptome	Biased	Partial	Expressed	Direct	No	Relative
Metabolome	No	No	No	Downstream	Direct	Relative
Absolute reference (qPCR / flow / spike-in)	Targeted	Targeted	No	No	No	Direct

Measurement



Direct



Partial / inferred



Not measured